$$(\sqrt{x-1}+1)^2 = x + 2\sqrt{x-1}$$

$$(\sqrt{x-1}-1)^2 = x - 2\sqrt{x-1}$$

(i)

$$\begin{aligned}\int_5^{10} \frac{\sqrt{x+2\sqrt{x-1}}+\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}}\,dx &= \int_5^{10} \frac{\sqrt{x-1}+1+\sqrt{x-1}-1}{\sqrt{x-1}}\,dx \\ &= \int_5^{10} 2\,dx \\ &= 10\end{aligned}$$

(ii) Let $u = \sqrt{x-1},\ x = \frac{5}{4} \implies u = \frac{1}{2},\ x = 10 \implies u = 3$

$$dx = 2u\,du$$

$$y = \frac{\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}} = \frac{|u-1|}{u}$$

$$\begin{aligned}\text{Area} &= \int_{\frac{1}{2}}^{1} \frac{1-u}{u}\cdot 2u\,du + \int_1^3 \frac{u-1}{u}\cdot 2u\,du \\ &= \int_{\frac{1}{2}}^{1} (2-2u)\,du + \int_1^3 (2u-2)\,du \\ &= \left[2u-u^2\right]_{\frac{1}{2}}^{1} + \left[u^2-2u\right]_1^3 \\ &= \frac{1}{4} + 4 \\ &= \frac{17}{4}\end{aligned}$$

(iii)

$$
\begin{aligned}
&\int_{\frac{5}{4}}^{10} \frac{\sqrt{x+2\sqrt{x-1}}+\sqrt{x-2\sqrt{x+1}+2}}{\sqrt{x^2-1}}\,dx \\
&= \int_{\frac{5}{4}}^{10} \frac{\sqrt{x-1}+1+\sqrt{x+1}-1}{\sqrt{x-1}\sqrt{x+1}}\,dx \\
&= \int_{\frac{5}{4}}^{10} \left(\frac{1}{\sqrt{x+1}}+\frac{1}{\sqrt{x-1}}\right)dx \\
&= \left[2\sqrt{x+1}+2\sqrt{x-1}\right]_{\frac{5}{4}}^{10} \\
&= 2\sqrt{11}+2
\end{aligned}
$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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